

PLOVDIV, Bulgaria

WMO: 156250

Lat: 42.068N

Lon: 24.851E

Elev: 597

StdP: 14.38

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	19.1	8.3	8.6	19.1	12.3	10.5	24.0	30.6	40.4	26.9	38.7	4.1	280	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	94.9	69.5	91.7	69.0	89.4	68.3	73.0	87.9	71.4	85.6	70.1	83.8	7.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	106.6	77.8	66.6	100.3	76.2	65.9	97.6	75.6	37.0	87.8	35.6	85.4	34.4	84.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.0	21.6	18.4	DB	7.0	100.4	6.7	4.8	2.1	103.9	-1.8	106.7	-5.6	109.4	-10.4	112.9	(4)
			WB	6.3	77.3	6.6	2.4	1.6	79.0	-2.2	80.4	-5.9	81.8	-10.7	83.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	34.2	38.7	46.3	54.8	63.7	71.4	75.9	75.6	67.1	56.4	46.2	36.5
	DBStd	16.00	8.09	8.44	7.33	6.16	5.42	5.33	4.48	4.32	5.75	6.65	7.15	8.08
	HDD50	1597	492	323	158	24	0	0	0	0	0	21	155	424
	HDD65	4447	956	737	579	309	89	10	0	0	43	274	565	885
	CDD50	3664	1	6	45	167	426	643	803	795	514	220	40	4
	CDD65	1041	0	0	0	3	50	203	339	330	107	8	1	0
	CDH74	11278	0	0	7	79	605	2080	3714	3596	1073	118	6	0
	CDH80	4728	0	0	1	11	129	819	1735	1660	356	17	0	0
Wind	WSAvg	6.4	6.0	6.7	7.4	6.9	6.6	6.8	7.0	6.2	6.1	5.4	5.3	5.8
Precipitation	PrecAvg	24.2	1.9	1.7	1.9	1.7	2.7	2.3	2.3	1.7	1.8	1.9	1.6	2.7
	PrecMax	34.1	4.3	4.5	5.5	3.2	6.8	4.7	6.7	4.7	5.0	4.4	5.0	5.2
	PrecMin	13.9	0.3	0.1	0.2	0.5	0.8	0.8	0.0	0.0	0.4	0.2	0.1	0.9
	PrecStd	5.1	1.0	1.1	1.2	0.9	1.5	1.1	1.8	1.3	1.2	1.1	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	66.4	73.7	81.0	87.6	95.2	100.7	98.7	92.2	82.4	73.4	62.8
		MCWB	48.4	51.3	54.8	61.7	67.0	70.2	69.5	69.3	67.0	64.1	57.0	51.6
	2%	DB	53.7	61.1	69.4	75.5	83.8	91.4	95.2	94.7	87.9	77.1	66.4	55.6
		MCWB	45.4	49.6	53.4	59.3	64.8	69.8	69.8	69.1	65.5	61.5	55.8	48.2
	5%	DB	49.3	56.8	64.6	71.6	80.5	88.0	91.7	91.6	84.2	73.2	61.2	50.3
		MCWB	44.0	47.8	51.4	57.1	63.9	68.1	69.0	68.9	64.7	59.8	53.8	45.5
	10%	DB	45.0	51.6	60.5	67.9	77.2	84.5	89.2	89.1	80.4	68.5	57.2	47.8
		MCWB	41.4	44.8	49.6	55.8	62.6	67.2	68.4	67.9	63.6	58.1	51.4	44.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	54.2	57.8	64.6	70.1	74.8	74.7	75.2	70.2	66.3	60.3	54.0
		MCDB	58.0	62.6	68.7	75.7	82.5	88.6	90.3	88.8	82.4	77.5	68.6	59.3
	2%	WB	46.9	50.7	54.5	61.1	67.5	72.2	72.7	73.1	68.2	63.3	57.0	49.5
		MCDB	51.8	58.7	65.7	71.9	79.1	86.4	87.4	88.3	81.1	73.2	64.7	53.4
	5%	WB	44.4	48.4	52.4	58.7	65.5	70.3	71.2	71.0	66.6	61.2	54.8	46.9
		MCDB	47.9	55.4	62.7	68.9	76.8	83.3	85.5	85.3	79.0	69.4	60.2	50.1
	10%	WB	41.7	45.6	50.2	56.7	63.7	68.7	70.0	69.5	65.2	59.4	52.5	44.4
		MCDB	44.7	50.3	58.9	65.9	74.0	81.1	84.3	83.4	76.7	66.7	56.6	46.6
Mean Daily Temperature Range	5% DB	MDBR	14.3	17.0	20.0	21.6	22.1	22.8	23.7	24.3	22.5	19.6	16.2	14.1
		MCDBR	21.7	26.7	29.4	28.5	28.1	28.1	29.1	29.0	29.1	27.3	23.8	20.3
		MCWBR	15.2	17.0	16.1	14.9	13.2	11.1	10.9	10.9	12.5	14.1	14.3	14.0
	5% WB	MCDBR	19.7	24.7	26.6	25.2	24.8	25.4	25.5	25.2	24.2	23.2	20.1	17.7
		MCWBR	14.6	16.4	15.3	14.4	13.0	11.4	10.6	11.0	11.6	13.1	12.7	12.8
Clear-Sky Solar Irradiance	taub	0.338	0.373	0.425	0.489	0.475	0.478	0.477	0.486	0.457	0.408	0.368	0.343	
	taud	2.338	2.235	2.117	1.954	2.070	2.107	2.123	2.090	2.132	2.251	2.322	2.329	
	Ebn at Noon	251	258	257	249	256	255	254	247	244	243	236	236	
	Edh at Noon	27	35	44	56	51	49	49	49	43	34	28	25	
All-Sky Solar Radiation	RadAvg	532	787	1114	1451	1772	1942	2041	1859	1382	932	609	464	
	RadStd	74	98	100	161	133	162	152	127	104	100	60	66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.88	N/A	N/A	N/A	N/A	N/A	-191	-274	N/A	N/A
Regional (28 neighbors)		+0.81	N/A	N/A	N/A	+1.10	+1.13	-189	-288	+128	N/A

Nomenclature: See separate page